

# Jaume A. Badia-Boher

*Postdoctoral Researcher in Statistical Ecology*

Swiss Ornithological Institute · Sempach, Switzerland

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## RESEARCH PROFILE

I develop and apply Bayesian statistical models to understand animal population dynamics and species distributions. My work spans spatially explicit demographic models, integrated population models, mark-recapture frameworks, and species distribution models, with a focus on translating demographic inference into evidence-based management.

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## EMPLOYMENT

### Postdoctoral Researcher in Statistical Ecology

*Sep 2023 – present*

*Swiss Ornithological Institute (Vogelwarte), Switzerland · Supervisors: Marc Kéry & Michael Schaub*

- Developing novel Bayesian integrated population models that estimate variation in demographic parameters across space.
- Evaluating the impact of emerging wildlife diseases on population dynamics and resilience.
- Generating demographic insights to support evidence-based management in species of conservation interest.
- Developing simulation-based frameworks to evaluate the performance of species distribution and population dynamics models.

### Doctoral Researcher in Biodiversity

*2020 – 2023*

*Universitat de Barcelona (UB) · Supervisor: Antonio Hernández-Matías*

- Thesis: *Population Models: Flexibility, Advances, and Applications to Wildlife Conservation – The Bonelli's Eagle as a Study Case*
- Developed novel Bayesian mark-recapture models; generated new insights for the conservation of long-lived species, with a focus on the Catalan and Mallorcan Bonelli's eagle populations.

### Red List Officer

*Jun – Sep 2017*

*IUCN Red List Unit, International Union for Conservation of Nature, Cambridge, UK*

- Contributed to species extinction risk assessments for the global IUCN Red List of Threatened Species

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## EDUCATION

### PhD in Biodiversity

*2020 – 2023*

*Universitat de Barcelona · [see Employment for details]*

- Honours in PhD Thesis
- PhD Extraordinary Award

### MSc in Biology – Animal Ecology

*2016 – 2018*

*Lund University, Sweden · Supervisors: Ana Sanz-Aguilar, J.A. Donazar, & Staffan Bensch*

- Thesis: *Evaluating European LIFE conservation projects: Improvements in survival of an endangered vulture.*
- Honors in MSc. Thesis.

- BSc. Top Mark Award.

## SELECTED SCIENTIFIC PUBLICATIONS

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\* = corresponding author | Citations as of June 2026: 128 | h-index: 4

1. [Badia-Boher JA\\*](#), Schaub M, Prommer M, Kéry M (2026). **Making survival spatial: an integrated model for territory occupancy and capture-recapture data**. In review at *Methods in Ecology and Evolution*. DOI: [10.32942/X2BH4D](#)
2. [Badia-Boher JA\\*](#), Schaub M, Mollet M, van Geneijgen P, van der Jeugd HP, Caliendo V, Kéry M (2026). **Evaluating the demographic impacts of the highly pathogenic avian influenza panzootic**. *Journal of Applied Ecology*, 63(1): e70234. DOI: [10.1111/1365-2664.70234](#)
3. Barczyk J, Kéry M, Neubauer G, Kellner KF, Soh MCK, [Badia-Boher JA](#) (2026). **Performance of generalized distance sampling models with temporary emigration: a simulation study**. *Oikos*, 2026(2): e11544. DOI: [10.1002/oik.11544](#)
4. [Badia-Boher JA\\*](#), Hernández-Matías A, Mañosa S, Parés F, Bas JM, Arévalo-Ayala DJ, Real J (2025). **Fall and rise of a threatened raptor: Unraveling long-term population dynamics with spatially explicit integrated models**. *Ecological Applications*, 35(2): e70013. DOI: [10.1002/eap.70013](#)
5. [Badia-Boher JA\\*](#), Real J, Riera JL, Bartumeus F, Parés F, Bas JM, Hernández-Matías A (2023). **Joint estimation of survival and dispersal effectively corrects the permanent emigration bias in mark-recapture analyses**. *Scientific Reports*, 13(1): 6970. DOI: [10.1038/s41598-023-32866-0](#)
6. [Badia-Boher JA\\*](#), Sanz-Aguilar A, De la Riva M, Gangoso L, et al. (2019). **Evaluating European LIFE conservation projects: Improvements in survival of an endangered vulture**. *Journal of Applied Ecology*, 56(5): 1210–1219. DOI: [10.1111/1365-2664.13350](#)

## FULL PUBLICATION LIST

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### Journal articles

7. Arbeiter S, Morkvénas Ž, [Badia-Boher JA](#), Oppel S (2026). **Reinforcement is not enough to establish a viable population of the globally threatened *Acrocephalus paludicola* (Aquatic Warbler) at the margin of its breeding range**. *Ornithological Applications*, duag012. DOI: [10.1093/ornithapp/duag012](#)
8. Saggese MD, Falk K, Zuberogoitia I, Sánchez R, Ellis DH, [Badia-Boher JA](#), et al. (2026). **Peregrine Falcon Populations Under Threat Again**. *Animal Conservation*, 29(2): 133–135. DOI: [10.1111/acv.70050](#)
9. Schaub M, [Badia-Boher JA](#) (2025). **Comparison of Bayesian Models to Estimate Survival From Dead-Recovery Alone and Together With Live-Encounter Data: Challenges and Opportunities**. *Ecology and Evolution*, 15(6): e71517. DOI: [10.1002/ece3.71517](#)
10. Arévalo-Ayala DJ, Real J, Margalida A, [Badia-Boher JA](#), Mañosa S, et al. (2025). **Contrasting vital rate contributions across interconnected populations of a highly vagile avian scavenger: A multisite modelling approach**. *Biological Conservation*, 311: 111454. DOI: [10.1016/j.biocon.2025.111454](#)
11. [Badia-Boher JA\\*](#), Real J, Hernández-Matías A (2024). **Assumptions about survival estimates and dispersal processes can have severe impacts on population viability assessments**. *Biological Conservation*, 292: 110550. DOI: [10.1016/j.biocon.2024.110550](#)
12. [Badia-Boher JA\\*](#), Hernández-Matías A, Viada C, Real J (2022). **Raptor reintroductions: Cost-effective alternatives to captive breeding**. *Animal Conservation*, 25(2): 170–181. DOI: [10.1111/acv.12729](#)

## Preprints & manuscripts under review

13. Prommer M, Badia-Boher JA, Kéry M, Schaub M, Bagyura J, Oli MK (2026). **Evaluating population resilience to anticipated stressors using integrated population modeling: a case study of Peregrine Falcons**. In review at *Journal of Ornithology*. DOI: [10.32942/X2BH31](https://doi.org/10.32942/X2BH31)
14. Kéry M, Monneret RJ, Badia-Boher JA, Schaub M (2025). **Demographic causes of the pesticide crash in the peregrine falcon**. In review at *Ecology*. DOI: [10.32942/X2R05B](https://doi.org/10.32942/X2R05B)
15. Muller M, Badia-Boher JA (2026). **A new computational framework for speeding up the fitting of multistate capture-mark-recapture models**. Submitted to *Methods in Ecology and Evolution*. DOI: [10.32942/X2539M](https://doi.org/10.32942/X2539M)

## TECHNICAL REPORTS

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16. [Badia-Boher JA](#), Kéry M, Strebel N (2024). **Power analysis for a potential white-backed woodpecker monitoring design in Switzerland**. *Swiss Ornithological Institute*.
17. Hernández-Matías A, [Badia-Boher JA](#), Real, J (2021). **Estudio sobre la viabilidad de la extracción de pollos en la población de águila de Bonelli *Aquila fasciata* en la isla de Mallorca**. *Universitat de Barcelona & Aquila A-LIFE*.

## QUANTITATIVE SKILLS

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Numbers in brackets refer to the numbered publication list above.

- **Development of new integrated modelling approaches to study spatial heterogeneity in population dynamics** [1,4,5]
- **Bayesian hierarchical modelling and computation:** BUGS-language MCMC inference engines (JAGS, NIMBLE) [1,2,4,5,6,7,9,10,11,13,14,15]
- **Integrated population models (IPMs):** joint study of survival, productivity, and population dynamics from the integration of different ecological data sources [2,4,7,10,13,14]
- **Mark-Recapture (CMR) models:** analysis of single-state, multi-state, dispersal, and dead-recovery data [5,6,9,11,12,15]
- **Species Distribution Models:** single-state, multistate, and dynamic occupancy models, power analyses and integration with other data sources [1,16]
- **Distance Sampling models:** simple and generalized hierarchical distance sampling models, simulation-based models and assessments of abundance dynamics [3]
- **Model optimization and simulation:** likelihood customization, MCMC sampler tuning, simulation studies to evaluate model performance [9,15]
- **Population Forecasting:** demographic resilience evaluations and viability analyses [2,7,11,12,13]

## AWARDS, HONOURS, & GRANTS

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| <b>1st Best Oral Presentation – SymbioSE 23</b><br><i>Symposium of Biology Students in Europe, Koper, Slovenia</i>                                | 2023 |
| <b>1st Best Student Poster – EURING Analytical Meeting &amp; Workshop</b><br><i>EURING, Québec City, Canada</i>                                   | 2021 |
| <b>Shortlisted – Southwood Prize Early Career Researcher Award</b><br><i>British Ecological Society, London, UK</i>                               | 2020 |
| <b>Doctoral Grant – Ajuts Joan Oró</b><br><i>Agència de Gestió d'Ajuts Universitaris i de Recerca (AGAUR), Generalitat de Catalunya · €72.138</i> | 2020 |

## SELECTED PRESENTATIONS

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- Badia-Boher JA (2026). *Cap a una perspectiva espacial en l'estudi demogràfic dels ocells europeus: reptes, oportunitats i dades*. Institut Català d'Ornitologia (ICO), Barcelona, Spain. *[Invited talk]*
- Badia-Boher JA (2026). *Computational efficiency in Bayesian mark-recapture models*. Population Ecology Seminar Series, University of Zurich, Zurich, Switzerland. *[Invited talk]*
- Badia-Boher JA, Mollet M, van Geneijgen P, Schaub M, van der Jeugd H, Kéry M (2025). *First insights on the effect of recent Avian Influenza outbreaks on long-lived bird populations*. 15th Meeting of the European Ornithologists' Union (EOU), Bangor, Wales. *[Contributed talk]*
- Badia-Boher JA (2025). *Survival Estimation and Population Viability Analysis*. Congress of the European Bird Census Council (EBCC), Riga, Latvia. *[Contributed talk]*

## WORKSHOP TEACHING AND SYMPOSIUM ORGANIZATION

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- Kéry M, Schaub M, Badia-Boher JA, Ketwaroo FR, Muller MH (2025). *Bayesian population analysis using NIMBLE and JAGS* (Workshop – 40 hrs). Swiss Ornithological Institute, Sempach, Switzerland.
- Badia-Boher JA, Martin EC, Paquet M (2025). *Two decades in review – advances in Integrated Population Models and applications to bird population dynamics studies* (Symposium – 4 hrs). 15th Meeting of the European Ornithologists' Union, Bangor, Wales.
- Badia-Boher JA (2024). *Introduction to Capture-Mark-Recapture Analysis in RStudio* (Workshop – 8 hrs). 1st EUFLYNET Meeting & Training School, Jastarnia, Poland.

## UNIVERSITY TEACHING

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### Faculty of Biology – Universitat de Barcelona

2021 – 2022

- Teaching Assistant: *Population Viability Analysis* (MSc in Biodiversity).
- Teaching Assistant: *Conservation Biology* (BSc in Environmental Science).

## RESEARCH SUPERVISION AND MENTORING

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### Swiss Ornithological Institute

2023 – present

- PhD Committee Member for Daniel Breitenmoser (University of Zurich, 2026–2029): *Developing a framework for evidence-based decision making in wild ungulates*.
- PhD Thesis Chapter Mentor for Julia M. Barczyk (University of Wrocław, 2023–2025): *Use of Hierarchical Distance Sampling analyses to study avian biodiversity at the Białowieża forest*.
- Research Stay Mentor for Oriol Solà (CTFC, 2025): *Dealing with the dirty little secret of citizen science biodiversity monitoring: preferential sampling*.
- PhD Tribunal Member for Oriol Solà (CTFC, 2025): *Advancing Species Distribution Dynamics Modelling: Spatial Processes, Sampling Design, and Data Integration*.

## ACADEMIC SERVICE

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- Peer reviewer for: *Methods in Ecology and Evolution*, *Journal of Applied Ecology*, *Ecological Applications*, *Journal of Animal Ecology*, *Journal of Zoology*, *Journal of Raptor Research*, *Journal of Avian Biology*, *Ibis*, *Bird Conservation International*, *Animal Biodiversity and Conservation*.

## MEDIA COVERAGE

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- *Versió RAC1*, RAC1 (Radio). 8<sup>th</sup> September 2022 – On Catalonia's wild boar pest situation and management options [[tinyurl.com/j1rac1](https://tinyurl.com/j1rac1)].
- *Notícies Migdia*, Betevé (TV). 6<sup>th</sup> September 2022 – On the complexity of the wild boar pest around Barcelona city [[tinyurl.com/jbbtv1](https://tinyurl.com/jbbtv1)].
- *Adolescents XL*, Catalunya Ràdio (Radio), 25<sup>th</sup> May 2022 – On the status and management of wolf populations in Sweden and Catalonia.

## LANGUAGES

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- Catalan – native
- Spanish – native
- English – professional proficiency
- German – A2